Extracting High-Frequency Monetary Policy Expectations from Decentralized Prediction Markets

ABSTRACT

This report develops a methodology to extract market-implied monetary policy expectations from decentralized prediction markets. Using Polymarket contracts on Federal Open Market Committee (FOMC) rate decisions (April 2023 to April 2026), discrete outcome probabilities are transformed into a continuous "shadow rate" by fitting parametric distributions to bin-based market prices. Gaussian and Student-t models are estimated via CDF optimization, producing time series of expected policy rates and uncertainty. AIC-based model comparison indicates that the Student-t distribution provides a superior fit relative to the Gaussian specification. Extending the approach across multiple contracts allows construction of a forward curve, whose dynamics are analyzed using Principal Component Analysis and shown to exhibit Level, Slope, and Curvature factors consistent with traditional yield-curve theory. Compared with CME Federal Funds futures, prediction-market-implied rates respond more rapidly to new information and retain higher-order distributional features. These results suggest decentralized prediction markets provide a high-frequency, event-targeted measure of monetary policy expectations suitable for quantitative macro-financial applications.

I. INTRODUCTION

Monetary policy expectations play a central role in financial markets, influencing asset prices well before official policy decisions are announced. In most data science applications involving macroeconomic variables, the Federal Reserve policy rate is typically incorporated as a stepwise constant time series, where the rate remains unchanged between Federal Open Market Committee (FOMC) meetings and is forward-filled until the next announcement. While this representation reflects the realized policy rate, it fails to capture the evolving expectations of market participants in the period leading up to policy decisions.

Financial markets, however, continuously adjust to anticipated monetary policy changes. Expectations regarding future interest rate movements are incorporated into asset prices through bonds, equities, currencies, and derivatives. As a result, using only the realized policy rate may introduce a lag in predictive models, particularly in applications such as market regime detection (as was the case in our group project) or macro-driven asset allocation. A forward-filled rate assumes that policy expectations remain constant between meetings, ignoring the continuous updating of expectations embedded in market prices.

This report proposes a framework to extract continuous market-implied policy rate distributions from Polymarket contracts (referred to as "the shadow rate" in this report)[1], construct a forward curve of expectations, and compare the resulting signal with CME Federal Funds futures.

II. METHODOLOGY & DATA

A. Polymarket contracts

Polymarket is a decentralized, blockchain-based prediction market platform built on the Polygon network (an Ethereum Layer-2 scaling solution). It allows participants to trade shares in the outcomes of future real-world events, ranging from macroeconomic policy decisions to electoral results. Structurally, Polymarket operates as a Central Limit Order Book (CLOB) exchange where outcomes are tokenized into binary options. Prediction markets like Polymarket aggregate dispersed private information into a single public metric: price. As demonstrated in [1], these markets effectively act as sufficient statistics, efficiently pricing in all available public and dispersed human-held information Because shares for a specific outcome (almost) always resolve to either $1.00 (if the event occurs) or $0.00 (if it does not), the trading price of a share at any given time t theoretically represents the market's risk-neutral probability estimation that the event will occur. Thus, if a share representing a "25 basis point rate cut" trades at $0.35, the market implies a 35% probability of that outcome, assuming risk neutrality among participants and negligible transaction frictions.

Formally, a Polymarket event partitions the state space Ω into a set of n mutually exclu-

[1] This terminology should not be confused with the Shadow-Rate Term Structure Models (SRTSMs) commonly used in macroeconomics (e.g. Wu and Xia (2016) or Krippner (2013)) which represent the stance of monetary policy when the target rate is constrained by the zero lower bound.

sive and collectively exhaustive discrete events, $E = \{E_1, E_2, \ldots, E_n\}$, such that:

$$\bigcup_{i=1}^{n} E_i = \Omega \quad \text{and} \quad E_i \cap E_j = \emptyset \text{ for } i \neq j$$

For each event E_i (e.g., "rate cut of 25 bps"), the market issues a binary derivative contract. The terminal payoff function $X_i(\omega)$ of this contract at maturity T is governed by an indicator function:

$$X_i(\omega) = \mathbb{I}_{E_i}(\omega) = \begin{cases} 1 & \text{if the realized state } \omega \in E_i \\ 0 & \text{if the realized state } \omega \notin E_i \end{cases}$$

Let the market be defined over a finite time horizon $t \in [0, T]$, where T is the resolution date of the event (e.g., the conclusion of the FOMC meeting).

At any time $t < T$, the contract for event E_i trades at a market price $\pi_t(E_i)$. By the First Fundamental Theorem of Asset Pricing, assuming the market is free of arbitrage, there exists an equivalent martingale measure $\mathbb{Q}$ (the Risk-Neutral Measure) such that the price of any asset is the expected discounted value of its future payoff. Assuming a risk-free interest rate of $r = 0$ (which is standard in decentralized prediction markets where capital is locked in non-yield-bearing stablecoins like USDC during the trade), the price of the contract is the conditional expectation of its payoff under $\mathbb{Q}$ and $\mathcal{F}_t$ the market filtration [2]:

$$\pi_t(E_i) = \mathbb{E}^{\mathbb{Q}}[X_i(\omega) \mid \mathcal{F}_t] = \mathbb{Q}(E_i \mid \mathcal{F}_t)$$

It is critical to note that the market price $\pi_t(E_i)$ extracts the risk-neutral probability $\mathbb{Q}(E_i \mid \mathcal{F}_t)$, not the true physical probability $\mathbb{P}(E_i \mid \mathcal{F}_t)$. $\mathbb{Q}$ incorporates the aggregate risk aversion of the market participants. However, prediction markets differ structurally from traditional derivatives markets. Contracts are fully collateralized, payoffs are binary and relatively small, and participants are predominantly speculative rather than hedging institutional balance sheet risk. As a result, the risk premium embedded in prices is expected to be limited. Furthermore, FOMC outcomes are not strongly correlated with typical portfolio losses, reducing the demand for risk-transfer. Under these conditions, the wedge between risk-neutral and physical probabilities is likely small, allowing market prices to serve as a reasonable proxy for the physical probability distribution [3]:

$$\pi_t(E_i) \approx \mathbb{P}(E_i \mid \mathcal{F}_t)$$

B. Federal Funds Rate Markets

The specific markets analyzed in this study involve the Federal Open Market Committee's (FOMC) decisions regarding the target federal funds rate. Let R_0 denote the upper bound of the target federal funds rate prior to a scheduled FOMC meeting, and let R_T denote the finalized upper bound immediately following the meeting at time T. The change in the rate is defined as $\Delta R = R_T - R_0$. Polymarket partitions the continuous sample space of potential rate changes into a set of discrete, mutually exclusive bins, denoted as $B = \{B_1, B_2, \ldots, B_n\}$. Each bin B_i is defined by a lower bound l_i and an upper bound u_i, such that $B_i = [l_i, u_i)$. For each bin B_i, a corresponding binary token is minted. The terminal payoff function $V(B_i)$ for holding one unit of this token is an indicator function:

$$V(B_i) = \begin{cases} 1 & \text{if } \Delta R \in B_i \\ 0 & \text{otherwise} \end{cases}$$

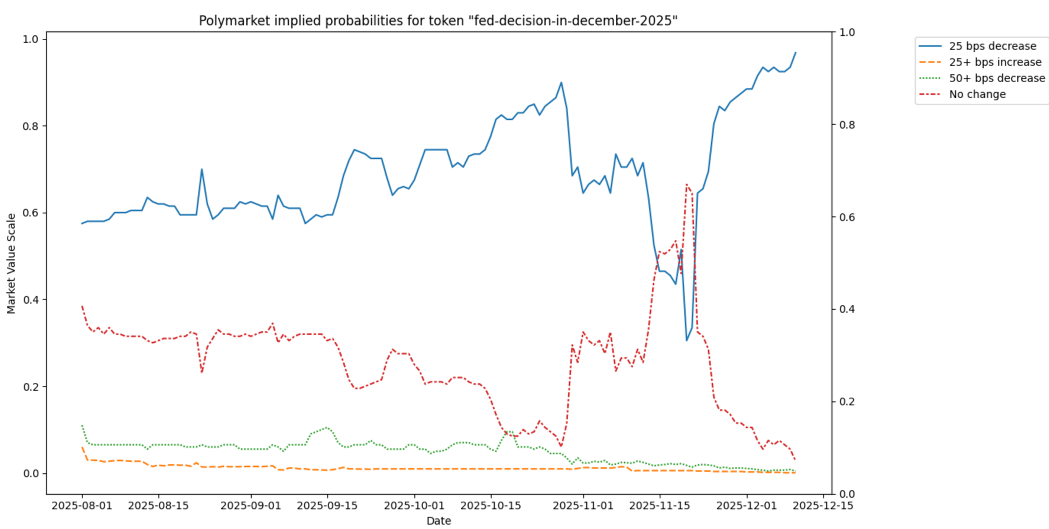


FIG. 1. Polymarket probabilities for the December 2025 decision contract

Constructed over a three-year period (April 2023 to April 2026), this data synthesizes on-chain prediction market activity with macroeconomic data. Earlier Polymarket data was omitted as contract rules were not consistent and low trading volumes prevented the accurate reflection of implied probabilities. In contrast, recent contracts exhibit deep liquidity. For instance, the March 2026 contract recorded a cumulative trading volume exceeding $260 million, ensuring robust and efficient price discovery. The Data was collected using:

- Macroeconomic Baseline (FRED): The daily historical baseline for the upper limit of the target federal funds range was sourced from the Federal Reserve Economic Data (FRED) API.

- Market Metadata (Gamma API): Polymarket's Gamma API was queried to identify relevant historical event slugs (e.g. fed-decision-in-december) and extract the underlying market metadata.

- Time-Series Pricing (CLOB API): Using the extracted Token IDs, Polymarket's Central Limit Order Book (CLOB) API was queried to download the daily closing prices ($p_{i,t}$) for each token over its active trading lifespan.

A critical microstructural feature of these markets is the evolution of their contract settlement rules, which necessitates different mathematical approaches:

- Regime 1: Strict Inequalities (Prior to May 2025 Contracts): Tokens were defined by strict boundaries. For example, a "25 bps decrease" token required a cut strictly greater than 0 but up to -25 bps. Mathematically, the bin bounds were defined using a minimal epsilon ($\epsilon \to 0^+$), such that $B = [-0.25, -\epsilon]$.

- Regime 2: Nearest-Neighbor Rounding (May 2025 Contracts and Beyond): The settlement methodology shifted to round any off-cycle rate change to the nearest 25 basis point standard increment. A "25 bps decrease" token therefore acts as a mathematically centered bin with a width of 25 basis points: $B = [-0.375, -0.125)$.

C. Probability density function estimation

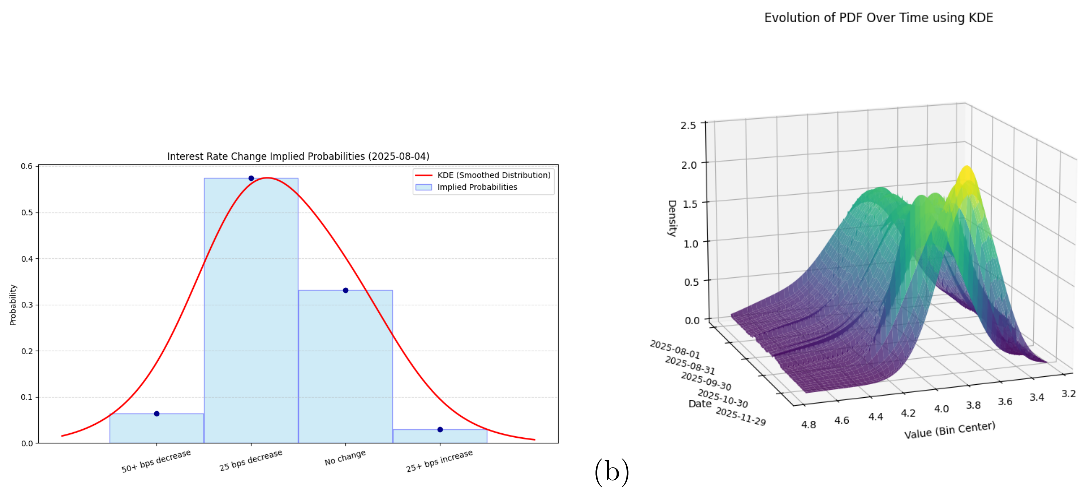
Macroeconomic pressure shifts continuously. The "true" interest rate the economy needs is a continuous variable. However, the Fed acts as a step-function filter. They observe this continuous "shadow rate" and round it to the nearest 25 bps peg. The goal is to model a continuous PDF to the implied probabilities in order to estimate this "shadow rate".

A first approach is to use Kernel Density Estimation (KDE): For a set of n data points $(x_1, x_2, ..., x_n)$, the KDE estimate at any point x is given by:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

where x is the location of the estimation, x_i are the data points, K is the kernel, a non-negative function that integrates to 1 and h the bandwidth, a smoothing parameter that controls the width of the kernel. As we already have discrete bins, the KDE treats the bin centers as the x_i coordinates and uses probabilities as weights. Instead of each point contributing $\frac{1}{n}$ to the height, each point contributes its specific probability weight to the sum.

(a)



(b)

FIG. 2. (a) KDE estimation on a single day snapshot, (b) evolution of the KDE estimation over the entire contract length

In the literature, it is often assumed that interest rates follow a normal distribution (Vasicek model) or log normal to prevent negative rates. Furthermore, KDE assigns probability based strictly on observed data point. It drops to zero very quickly outside the range of your observed market strikes therefore it does not extrapolate well.

D. Fitting continuous parametric distributions

A parametric distributional approach is preferred over non-parametric alternatives for several reasons. First, the number of observed probability bins is typically small, making fully non-parametric estimation unstable. Second, parametric distributions provide a smooth extrapolation beyond observed strikes, which is particularly important in the presence of unbounded tail bins. Third, the Gaussian distribution represents the maximum entropy distribution conditional on mean and variance, making it a natural baseline assumption. Finally, the parameters μ and σ admit direct economic interpretation as the market-implied policy rate and uncertainty. To better capture fat-tailed behavior frequently observed in financial markets, we extend the framework to a Student-t distribution, which introduces an additional degrees-of-freedom parameter controlling tail thickness.

A significant structural challenge in the raw Polymarket dataset is temporal overlap. The platform concurrently lists derivative contracts for multiple future FOMC meetings (e.g., trading on the January, March, and May decisions simultaneously during December). Consequently, cross-contract data inherently contains overlapping forward horizons. To construct a continuous time series, we isolate the "front-month" contracts. By filtering the dataset to include exclusively the immediate upcoming FOMC decision, we eliminate the complexities of path-dependency and compound probabilities. This simplification yields a single and non-overlapping timeline of short-term rate expectations.

The state space of potential rate targets is partitioned by the market into discrete bins. For any given contract, interior bins possess a uniform width—typically spanning a standard 25 basis point interval (e.g. ± 12.5 bps around a central target rate for regime 2). Conversely, the extreme boundary bins are unbounded to capture all tail risks. For example, a "+25 bps increase" token acts as an upper-bound tail, resolving to "Yes" if the interest rate increases by 25 basis points or more. As illustrated in Figure II C, a representative daily partition defines the following integration bounds:

$$B = \{(-\infty, 3.625),\ [3.625, 3.875),\ [3.875, 4.125),\ [4.125, +\infty)\}$$

Standard discrete expectation calculations (such as calculating the center of mass via a weighted average) fail structurally when applied to unbounded tail bins. To account for the infinite boundaries, we employ a Cumulative Distribution Function (CDF) optimization approach.

1. Gaussian distribution

We hypothesize that these probabilities are generated by a continuous Normal distribution with parameters $\theta = (\mu, \sigma)$. The theoretical probability $\hat{p}_i(\theta)$ that the realized rate falls within bin B_i is calculated using the Normal CDF, $F(x; \mu, \sigma)$:

$$\hat{p}_i(\mu, \sigma) = F(u_i; \mu, \sigma) - F(l_i; \mu, \sigma)$$

To estimate the optimal parameters $\hat{\mu}_t$ and $\hat{\sigma}_t$, we define a loss function $\mathcal{L}$ that minimizes the Sum of Squared Errors (SSE) between the empirical market probabilities and the theoretical CDF probabilities across all bins n:

$$\min_{\mu,\sigma} \mathcal{L}(\mu, \sigma) = \sum_{i=1}^{n} \left(\hat{p}_i(\mu, \sigma) - p_{i,t}\right)^2$$

Due to the non-linear nature of the CDF (it does not admit a closed form solution), optimizing this loss function requires numerical methods. We therefore estimate (μ, σ) numerically using the L-BFGS-B algorithm, a quasi-Newton optimization method that approximates the inverse Hessian while allowing for explicit parameter bounds. To help the algorithm converges successfully and avoids local minima, we initialize the optimizer with a heuristic prior (x_0). This prior is calculated by temporarily mapping the infinite tails to artificial midpoints (shifting them by 12.5 bps) and computing the discrete weighted average. The optimizer then refines this initial guess, allowing the tails to stretch to infinity naturally and extracting the true continuous parameters implied by the discrete market order book.

Under the "Regime 1" contract specifications, the "No Change" token presents a unique numerical challenge. Because this specific contract resolves exclusively on an exact 0 basis point change, its theoretical bin width was effectively zero. Attempting to fit a continuous Gaussian distribution into a zero-width bin causes the optimization algorithm to fail, as it attempts to collapse the entire probability mass into a single point. To resolve this, we relaxed the strict zero-width constraint for the "No Change" bin by introducing a symmetric expansion factor ($\epsilon = \pm 6.25$ bps). This granted the optimizer sufficient mathematical tolerance to fit the continuous density function across the discrete data without artificially driving the variance to zero.

Furthermore, while prediction markets aggregate information efficiently, they are occasionally subject to transient microstructure noise (e.g. temporary illiquidity or sudden order book sweeps). To extract a robust macroeconomic signal, we apply an Exponentially Weighted Moving Average (EWMA) to the daily optimized parameter estimates, $\theta_t \in \{\hat{\mu}_t, \hat{\sigma}_t\}$. The filtered estimates, $\tilde{\theta}_t$, are updated recursively:

$$\tilde{\theta}_t = \alpha\theta_t + (1 - \alpha)\tilde{\theta}_{t-1}$$

where the smoothing constant α is defined by the span N as $\alpha = \frac{2}{N+1}$. For this analysis, we set $N = 5$ days (corresponding to a trading week) providing a balance that dampens high-frequency variance while preserving the model's responsiveness to genuine macroeconomic data releases.

We can construct a continuous surface of market-implied rates and their associated uncertainty over time. As illustrated in Figure 3, the market-implied mean ($\tilde{\mu}_t$) serves as a high-frequency leading indicator. Rather than acting as a static forecast, the distribution dynamically adjusts in real-time, accurately reflecting shifts in aggregate market sentiment in advance of the actual FOMC policy rate decisions.

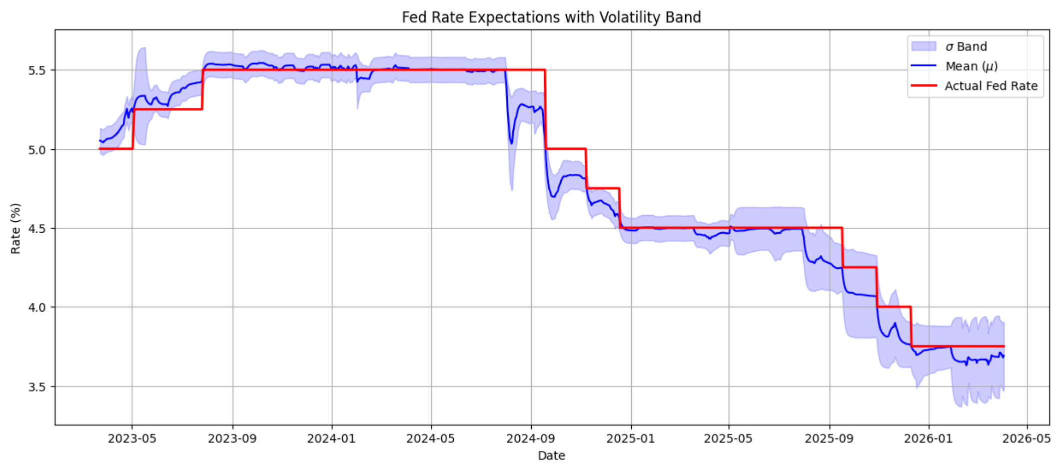


FIG. 3. Time series of implied rates using gaussian distribution

2. Student-t distribution

Gaussian distribution tends to underestimate tail risk. Financial markets tend to be fat tailed, a more appropriate distribution is the student t, by estimating a third parameter ν we can model tail behaviour. The modeling approach remains the same as in the previous section. We artificially limit the degrees of freedom to be greater than 3 to ensure the variance is defined.

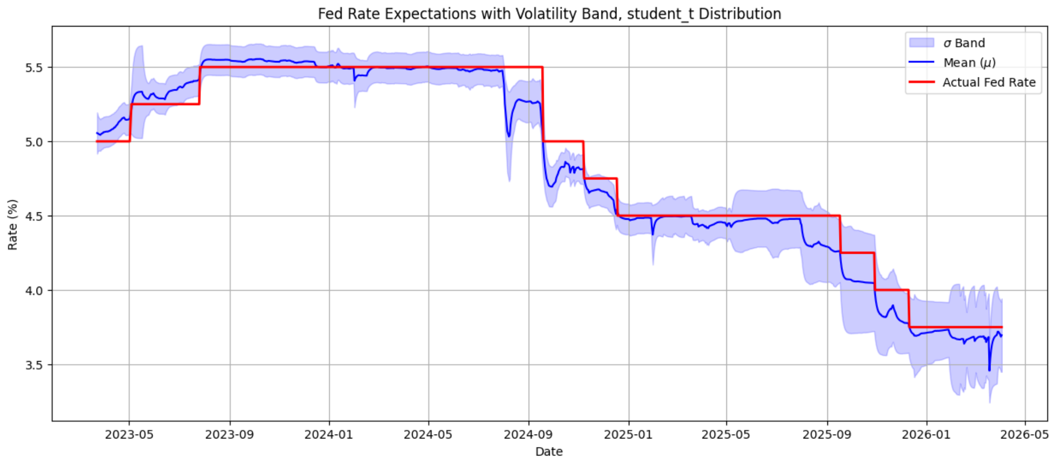


FIG. 4. Time series of implied rates using student-t distribution

Because the observed data consists of probabilities over finite bins rather than continuous density observations, the distribution parameters are only partially identified. Multiple parametric distributions can produce similar integrated probabilities across bins, particularly when the number of intervals is small. Hence , different parametric families such as Gaussian and Student-t may yield slightly different estimates of μ while providing comparable fits to observed probabilities (see Figures 3 and 4).

3. Model Comparison and Fat-Tail Risk

To evaluate whether the empirical data justified the addition of the degrees-of-freedom parameter (ν) to capture kurtosis, we utilized the Akaike Information Criterion (AIC). The AIC is a well-established econometric metric that estimates the relative quality of statistical models by balancing goodness-of-fit against model complexity. Formally defined as $AIC = 2k - 2\ln\left(\hat{L}\right)$—where k is the number of estimated parameters and $\hat{L}$ is the maximized value of the likelihood function. The criterion explicitly penalizes the addition of unnecessary variables to prevent overfitting. Therefore, a lower AIC score indicates a superior model.

The results demonstrate that the Student-t model significantly outperformed the standard Normal distribution, achieving a lower average AIC score across the sample **(-20.18 for the Student-t distribution vs. -18.02 for the Gaussian distribution)**. Because the AIC strictly penalizes parametric complexity (where the Student-t requires $k = 3$ parameters compared to the Gaussian's $k = 2$), this lower score mathematically proves that the improvement in the Sum of Squared Errors vastly outweighed the penalty of the third parameter.

Economically, this confirms that Polymarket participants actively price in "fat-tail" macroeconomic risks. It suggests that decentralized prediction markets dynamically allocate capital to the probability of extreme, non-standard monetary policy shocks that a traditional Gaussian framework would dismiss as statistically highly improbable.

E. Multi-contract forward curve

1. Fitting Gaussian Distribution to the Multi-Contract Forward Curve

While point-estimates of market-implied probabilities for the immediate FOMC cycle provide a valuable indicator, a robust macroeconomic projection requires the construction of a complete forward curve spanning multiple subsequent meetings $(T_1, T_2, \ldots, T_n)$. Since August 2024, the expansion of Polymarket's liquidity into longer-dated contracts has enabled the extraction of this term structure. However, a critical distinction must be made regarding the microstructure of these prediction markets: unlike terminal spot contracts, these derivatives function effectively as Forward Rate Agreements (FRAs). A contract maturing at T_n does not price the absolute Federal Funds Rate relative to the current spot $(t = 0)$. It reflects the expected marginal shift (ΔR_n) relative to the rate environment established at the penultimate meeting (T_{n-1}). To resolve the terminal rate from these marginal expectations, we apply a continuous probability density function (PDF) fit to the discrete price intervals of each meeting-specific contract. Under the principle of the Linearity of Expectation, the expected absolute target rate following future meeting n is derived as the sum of the current spot rate and the cumulative sequence of expected shifts:

$$\mathbb{E}_t[R_n] = R_0 + \sum_{k=1}^{n} \mathbb{E}_t[\Delta R_k]$$

2. Principal Component Analysis of the Implied Forward Term Structure

To test whether the decentralized prediction market accurately replicates the complex pricing dynamics of institutional fixed-income markets, we applied Principal Component

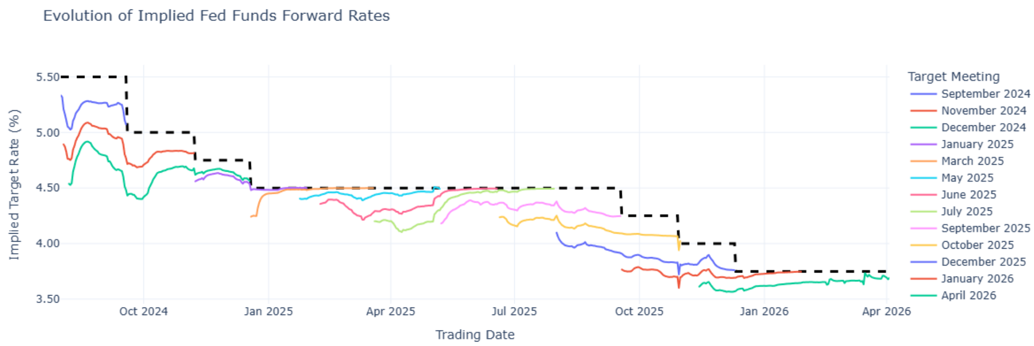


FIG. 5. forward implied rates (*True upper bound in dashed line)*

Analysis (PCA) to the derived term structure. This approach is rooted in the foundational empirical framework established by traditional financial econometrics.

In their 1991 paper [4], Robert Litterman and Jose Scheinkman demonstrated that the US Treasury yield curve is governed by a highly compressed latent dimensionality, despite consisting of dozens of interconnected maturities. By applying PCA to the daily changes of Treasury yields, they proved that over 95% of the curve's variance can be explained by just three orthogonal, unobservable macroeconomic factors:

- Level (Shift): The dominant factor (explaining 90% of variance), representing a parallel shift where all interest rates across the curve move up or down simultaneously in response to systemic macroeconomic shocks.

- Slope (Twist): The secondary factor, where short-term rates move inversely to long-term rates, causing the yield curve to steepen or flatten. This represents market expectations regarding the velocity or pace of monetary policy.

- Curvature (Butterfly): The tertiary factor, where medium-term rates move inversely to the extremes (short and long ends), causing the belly of the curve to bow. This captures idiosyncratic, maturity-specific supply and demand imbalances.

To determine if Polymarket participants implicitly trade this same multidimensional structure, we applied PCA to the daily stationary basis-point shifts $(\Delta\mu_{T1}, \Delta\mu_{T2}, \Delta\mu_{T3})$ of the smoothed implied forward curve. To prevent artificial volatility artifacts generated by contract expirations (the "roll effect"), differencing was strictly calculated on unique contract identifiers prior to constant-maturity alignment.

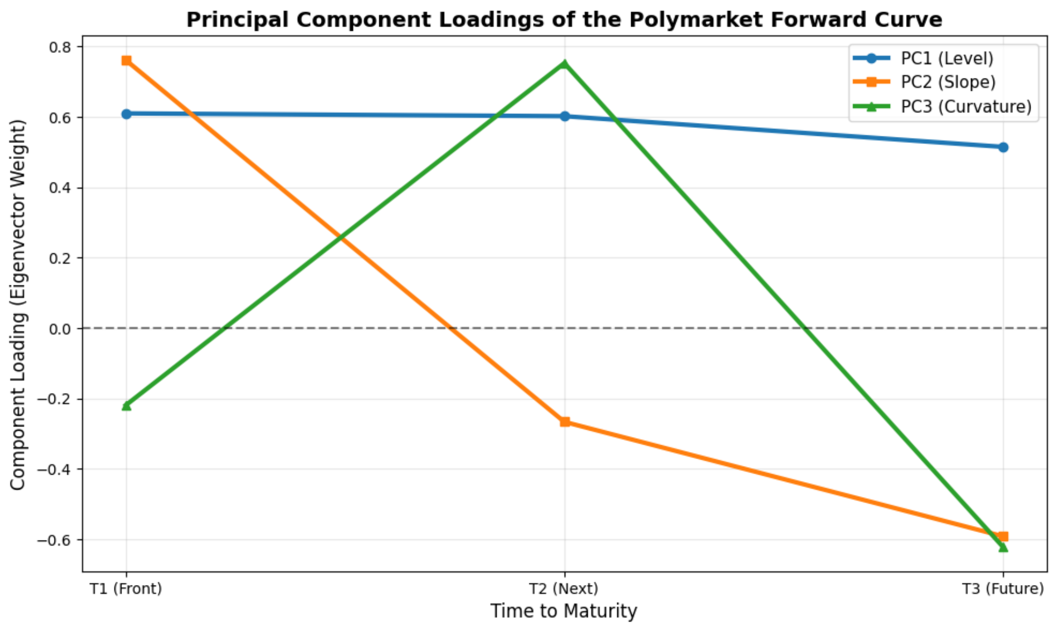


FIG. 6. PCA loading

The eigenvalue decomposition of the covariance matrix successfully extracted the classic three-factor "DNA" of the traditional yield curve. The extracted eigenvectors (component loadings) map perfectly to the theoretical Litterman-Scheinkman factors:

- **Principal Component 1 (The Level Factor), 81% of variance**: The empirical loading for PC1 emerged as a predominantly flat, horizontal vector across T_1, T_2, and T_3. Economically, this confirms that the vast majority of variance in Polymarket rate expectations is driven by systemic macroeconomic data (e.g. CPI or NFP releases). When inflation surprises the market, traders do not view sequential FOMC meetings as isolated events; they rationally shift the entire implied rate cycle in unison.

- **Principal Component 2 (The Slope Factor), 13% of variance:** The loading for PC2 exhibited a strict downward slope from T_1 to T_3. In linear algebra, eigenvectors are subject to sign ambiguity (i.e., v and $-v$ describe the exact same variance subspace). Therefore, this downward-sloping vector mathematically represents the "Flattening/Steepening" factor. It captures periods where the market debated the immediate pace of the Fed's actions (e.g. pivoting from a 25 bps cut to a 50 bps cut in the front-month (T_1)) while leaving the terminal, long-term destination (T_3) relatively anchored.

- **Principal Component 3 (The Curvature Factor), 6% of variance:** The loading for PC3 formed an inverse V-shape, peaking at T_2. Again reflecting eigenvector sign ambiguity, this is the classic "Butterfly" factor. It captures meeting-specific anomalies, such as the market pricing in a high probability of a "skip month" (e.g., cutting in March, pausing in May, and cutting again in July), which causes the middle of the curve to bow independently of the broader trend.

III. RESULTS

A. Traditional Finance (TradFi) Benchmarks vs. Decentralized Finance (DeFi) Prediction Markets

In traditional macroeconomic forecasting, the standard benchmark for extracting market-implied monetary policy expectations is the 30-Day Federal Funds Futures contract [5], traded on the Chicago Mercantile Exchange (CME). While highly liquid, the structural design of this derivative introduces specific econometric limitations when attempting to isolate high-frequency, event-driven policy signals.

The CME Federal Funds contract does not price the interest rate on a specific discrete day; rather, it settles against the arithmetic average of the daily Effective Federal Funds Rate (EFFR) for the applicable delivery month. The contract price (P) is quoted on an index basis, structurally defined as:

$$P = 100 - R_{implied}$$

Consequently, quantitative analysts back out the market's expectation of the average monthly rate by inverting the formula:

$$R_{implied} = 100 - P$$

If a contract expires in a month with D total days, and an FOMC meeting occurs on day d, the implied rate is mathematically constrained as the time-weighted average of the pre-meeting rate (r_0) and the expected post-meeting rate (r_1):

$$R_{implied} = \frac{d \cdot r_0 + (D - d) \cdot r_1}{D}$$

To accurately compare this implied effective rate against the Federal Reserve's target Upper Bound, we must adjust for the basis spread. Because the spread between the EFFR and the Upper Bound tends to remain structurally stable over short-to-medium-term horizons, we can approximate the implied Upper Bound by adding the empirical historical spread to $R_{implied}$.

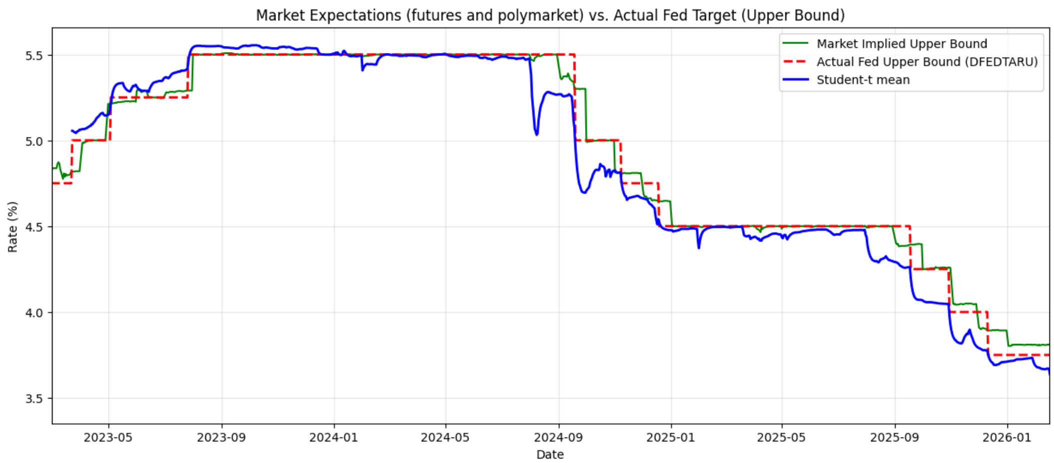


FIG. 7. CME Futures implied rates (green) vs. Polymarket implied rates (blue)

TABLE I. Comparison of Forecast Accuracy (difference between implied rate and FOMC outcome in bps)

Time to Event	**Student-t rate RMSE**	**CME rate RMSE**
0–3 Days	4.61	11.11
4–7 Days	6.75	12.45
8–14 Days	6.96	13.54
15–30 Days	8.23	18.08
31–60 Days	9.94	23.83

Figure 7 and Table I illustrate the empirical divergence between the CME futures implied rates and the prediction market implied rates. The data demonstrates that prediction markets offer a more responsive and accurate estimation of the expected target rate, whereas the futures-implied rate often exhibits temporal lag and rigidity to new information.

This divergence highlights three primary limitations of the TradFi benchmark that decentralized prediction markets natively resolve:

- **Temporal Distortion:** Because the Federal Reserve typically schedules FOMC meetings in the middle of the month, the front-month CME contract is subject to the "locked-in average" effect. If a rate cut occurs on the 15th of the month, the first 15 days permanently anchor the average to the older, higher rate. Consequently, the CME implied rate blends realized historical data with forward expectations, mathematically forcing it to lag behind the true post-meeting rate. By targeting specific discrete events, prediction markets completely bypass calendar-averaging artifacts.

- **Loss of Higher-Order Moments:** The pricing mechanism of a futures contract natively collapses market expectations into a single, one-dimensional point estimate. While $R_{implied}$ provides the first moment of the probability distribution (μ), it obscures all higher-order moments. In contrast, the continuous parametric estimation applied to Polymarket data successfully preserves variance (σ), skewness, and kurtosis, enabling rigorous stochastic modeling of uncertainty.

- **Institutional Risk Premia:** The CME Fed Funds market is dominated by institutional hedgers (e.g., commercial banks) utilizing the market to manage systemic balance sheet risk. Because these institutions are willing to pay a premium for hedging utility, $R_{implied}$ represents the true expected rate plus a time-varying term premium [6]. Fully collateralized prediction markets, primarily driven by speculative arbitrage rather than balance sheet constraints, offer a purer extraction of the underlying macroeconomic signal.

B. Density Calibration and Microstructure Collapse

To rigorously assess whether the continuous distributions extracted from Polymarket represent true physical probabilities ($\mathbb{P}$-measure) rather than risk-distorted expectations ($\mathbb{Q}$-measure), expanding on the hypothesis in II A, we evaluated the calibration of the implied densities using the Probability Integral Transform (PIT) framework established by Diebold, Gunther, and Tay (1998) [7] and the cumulative distribution function of the fitted Student-t distribution. If a sequence of density forecasts is optimal and correctly calibrated, the sequence of realized PITs will be independent and uniformly distributed on the unit interval $(0, 1)$. We tested the null hypothesis of uniformity across $N = 24$ independent FOMC meetings using the Kolmogorov-Smirnov (KS) and Cramér-von Mises (CvM) test statistics.

The empirical results reveal a distinct, horizon-dependent calibration effect. At a 28-day forward horizon, we fail to reject the null hypothesis of uniformity ($p_{KS} = 0.1176$, $p_{CvM} = 0.2566$). This confirms that at medium-term horizons, the decentralized prediction market is highly efficient; risk premia are negligible, and the implied density serves as an unbiased, mathematically sound physical density.

Conversely, at a 3-day forward horizon, uniformity is rejected at the 5% significance level ($p_{KS} = 0.0348$, $p_{CvM} = 0.0396$). Rather than indicating a sudden surge in risk premia, this near-term miscalibration is primarily driven by a mathematical breakdown we term "microstructure collapse." In the immediate days preceding an FOMC announcement, macroeconomic uncertainty largely evaporates, and market consensus typically approaches 100% certainty on a specific discrete target. Consequently, the probability mass collapses almost entirely into a single 25-bps bin. When our methodology attempts to fit a continuous Student-t density to this near-degenerate, single-point distribution, the parameter estimation becomes unstable. The continuous distribution is forced into an artificially narrow, highly kurtotic spike that fails standard continuous uniformity tests. The model is essentially underconfided as demonstrated by a humped PIT histogram.

While these calibration tests provide robust evidence for the medium-term validity of the methodology, it is important to note the econometric limitations of the sample size. The tests were constrained to $N = 24$ realized FOMC meetings. As non-parametric distribution tests like the KS and CvM statistics suffer from reduced statistical power in small samples, future research utilizing a larger multi-year dataset will be necessary to definitively map the exact temporal boundary where this microstructure collapse occurs.

IV. CONCLUSION & OUTLOOK

This study demonstrated a novel, data-driven methodology for extracting high-fidelity monetary policy expectations from decentralized prediction markets. By applying continuous parametric estimation to discrete, crowd-sourced probability mass functions, we successfully bypassed the structural limitations of traditional finance benchmarks. The resulting "Event-Targeted Shadow Rate" isolates pure macroeconomic sentiment, avoiding the temporal calendar averaging and institutional risk premia that distort the front-month CME

Federal Funds futures. Moreover, the application of PCA to the derived term structure confirmed alignment with the traditional Litterman-Scheinkman yield curve model. The autonomous emergence of orthogonal Level, Slope, and Curvature vectors from the Polymarket data proves that DeFi platforms process macroeconomic risk with an efficiency that mirrors institutional bond markets.

Despite these robust structural findings, it is imperative to acknowledge the nascent nature of the underlying data. Decentralized prediction markets have only achieved mainstream liquidity within the last few years. Consequently, the dataset utilized in this study spans approximately 3 years (24 FOMC meetings). In the context of macroeconomic econometrics, this is a fraction of a standard business cycle. Because the data does not yet span diverse monetary regimes (e.g prolonged zero-interest-rate policy era, severe recessionary crash, a multi-year hiking cycle) the long-term predictive validity and stability of these implied distributions remain empirically untested.

However, the mathematical framework established in this report lays the groundwork for a highly compelling future. As decentralized forecasting platforms continue to accumulate historical time series data and deepen in liquidity, the reliance on lagging, time-averaged traditional futures may become obsolete.

In the future, as the sample size expands across full macroeconomic cycles, this continuous Polymarket "Shadow Rate" could be integrated as a primary feature in advanced quantitative models. From dynamic asset pricing and algorithmic trading to dynamic risk-management algorithms, high-frequency, event-targeted probability distributions offer a structural evolution in how financial markets model the cost of capital. Ultimately, this approach proves that the "wisdom of the crowd," when mathematically structured, can successfully demystify the forward curve of global monetary policy.

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